



SALES PREDICTION USING PYTHON



CODSOFT DATA SCIENCE INTERNSHIP – TASK 4



OBJECTIVE

The objective of this project is to build a Simple Linear Regression model to predict product sales based on advertising data (TV, Radio, Newspaper).

This helps in understanding how advertising budget impacts sales performance.



PROJECT DESCRIPTION

This project uses the Advertising dataset which includes:



TV Advertising Budget



Radio Advertising Budget



Newspaper Advertising Budget



Sales (Target Variable)

The model mainly focuses on TV advertising vs Sales relationship using Simple Linear Regression.



TOOLS USED

Python 

Pandas

NumPy

Matplotlib 

Seaborn

Scikit-learn

VS Code 

FEATURES

- ✓ Load dataset using Pandas
- ✓ Data visualization using scatter plots
- ✓ Train-test split
- ✓ Model building using Linear Regression
- ✓ Prediction of sales
- ✓ Model evaluation

MODEL USED

Simple Linear Regression (OLS Method)

MODEL EQUATION

Sales = 6.94 + 0.0545 × TV

MODEL PERFORMANCE

Mean Absolute Error (MAE): 1.65

R² Score: 0.79

SAMPLE OUTPUT

- ✓ Dataset Head Displayed

✓ Actual vs Predicted Sales

✓ Regression Line Graph

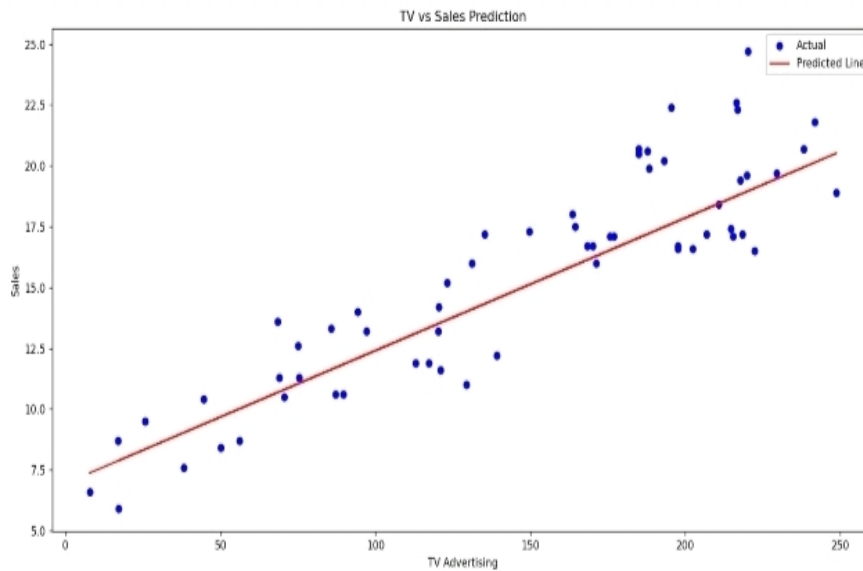
Dataset Head:

	TV	Radio	Newspaper	Sales
0	230.1	37.8	69.2	22.1
1	44.5	39.3	45.1	10.4
2	17.2	45.9	69.3	12.0
3	151.5	41.3	58.5	16.5
4	180.8	10.8	58.4	17.9

📊 Model Evaluation:
Mean Absolute Error: 1.65
R2 Score: 0.79

📌 Model Equation:
$$\text{Sales} = 6.94868320001357 + 0.054545752915907963 * \text{TV}$$

📊 GRAPH OUTPUT



✓ TV vs Sales Scatter Plot

✓ Regression Line Fit

CONCEPTS USED

- ✓ Data Handling (Pandas)
- ✓ Data Visualization
- ✓ Linear Regression
- ✓ Model Evaluation
- ✓ Train-Test Split

KEY LEARNINGS

- ✓ Understanding relationship between features and target
- ✓ Building regression models in Python
- ✓ Evaluating model accuracy
- ✓ Real-world prediction systems

CONCLUSION

The model successfully predicts sales based on TV advertising budget.

It shows a strong positive correlation between advertising spend and sales.

AUTHOR

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